AI CUSTOMER SUPPORT ASSISTANT

Performance Evaluation Report

Evaluation Date:	20 October 2026
Selected Evaluation Run:	`4760c90132b44e56a035d9ed6276b5c4`

1. Purpose

This evaluation measures three critical aspects of the application:

- Retrieval of relevant store-document evidence.
- Generated-answer correctness, groundedness, and complete response latency.
- Local application memory during idle operation, chat evaluation, and document processing.

Note: *Retrieval performance and generated-answer quality are assessed separately. Finding the expected evidence does not, by itself, demonstrate that the chatbot generated a correct answer.*

2. Test Environment

Component	Configuration
Computer	MacBook Air with Apple M1
Operating system	macOS 26.5.1
Architecture	ARM64
Physical memory	8 GiB
Python	3.12.11
Django	5.2.18
Web server	Local Django development server
Application address	http://127.0.0.1:8000
Document processing	Separate process_document_jobs worker
Chat model	gpt-4.1-mini

Embedding model	text-embedding-3-small
Filtered retrieval maximum distance	0.8

The selected chat evaluation used four version-1 demonstration documents: the product catalog, shipping policy, returns policy, and support policy. Each document had one indexed chunk. The corpus remained unchanged during that run.

3. Retrieval Evaluation

The baseline and filtered evaluations each contained 20 cases: 16 answerable or follow-up questions and four unsupported questions.

Evidence coverage measures whether the labeled supporting evidence appears among the retrieved results. Complete evidence rate measures whether all required evidence is present.

Metric	Baseline	Filtered
Cases completed without errors	**20/20**	**20/20**
Mean evidence coverage at k = 1, 3, and 5	**100%**	**100%**
Complete evidence rate at k = 1, 3, and 5	**100%**	**100%**
Unsupported cases returning matches	**4/4**	**0/4**
Unsupported no-match rate	**0%**	**100%**
Mean retrieval/rewrite evaluation latency	**756.10 ms**	**663.43 ms**

Filtering removed matches for all four unsupported questions while retaining the expected evidence for all 16 answerable and follow-up cases.

The filtered run's mean latency was 92.67 ms lower, approximately 12.3%. These individual runs do not establish a repeatable speed improvement: network and model-response variability may affect the measurement.

The 663.43 ms figure measures the retrieval/rewrite evaluation path, not complete chatbot response latency. These retrieval results are not "100% chatbot accuracy."

4. Complete Chat Evaluation

The selected run submitted 20 requests through the local chat HTTP endpoint and recorded actual generated answers, supporting citations, HTTP status, and response time.

The dataset contained:

- 12 answerable store questions.

- Four unsupported questions.
- Four follow-up questions using predefined conversation history.

Each follow-up's history was seeded before the measured request. This evaluates interpretation of supplied history rather than a complete, naturally generated conversation.

Review Method

Answers were manually reviewed against the expected answers and supplied source passages.

A correctness pass required the answer to meet the expected behavior without contradicting it. A groundedness pass required factual store claims to be supported by the cited passages. Unsupported cases passed when the assistant declined to supply an unrelated answer and explained the store-document limitation.

Summary Results

Measure	Result
Successful HTTP responses	**20/20 — 100%**
Request errors	**0**
Answerable questions passing manual review	**12/12**
Unsupported questions correctly declined	**4/4**
Follow-up questions passing manual review	**4/4**
Overall expected-behavior pass rate	**20/20 — 100%**
Answerable/follow-up responses with supporting citations	**16/16**
Answerable/follow-up responses passing groundedness review	**16/16**

The assistant correctly handled shipping thresholds, delivery estimates, return conditions, product specifications, and support-request requirements. It did not invent refund processing times or claim that a resolved ticket proved a refund had been paid.

Case-Level Review

Case	Topic	Correctness / Expected Behavior	Groundedness
E01	Shipping fee for SAR 180 subtotal	Pass	Pass
E02	Free shipping at exactly SAR 200	Pass	Pass

E03	Riyadh delivery estimate	Pass	Pass
E04	International shipping availability	Pass	Pass
E05	Shipping charge excluded from threshold	Pass	Pass
E06	Original packaging required for standard returns	Pass	Pass
E07	Defective personalized-item exception	Pass	Pass
E08	Damage-reporting deadline	Pass	Pass
E09	Unspecified refund processing time	Pass	Pass
E10	Lamp brightness limitation	Pass	Pass
E11	Tote bag dimensions	Pass	Pass
E12	Support-request contact requirements	Pass	Pass
E13	Football question	Pass: declined	Not applicable
E14	Baking question	Pass: declined	Not applicable
E15	Matrix determinant question	Pass: declined	Not applicable
E16	Mars question	Pass: declined	Not applicable
E17	Dammam delivery follow-up	Pass	Pass
E18	Lamp battery follow-up	Pass	Pass
E19	Personalized-name length follow-up	Pass	Pass
E20	Resolved ticket and refund-payment follow-up	Pass	Pass

Actual answer text and citation snapshots are retained in:
`chat_eval_20261020T162217259609Z_4760c901.json`
The JSON's answer_quality_review field remains pending; the judgments above reflect the subsequent manual review.

5. Complete Response Latency

Latency was measured from the local HTTP POST through receipt of the complete response body. It includes application processing and remote API waits, but excludes setup, history seeding, and browser rendering.

Metric	Result

Mean	**4.66 seconds**
Median	**4.19 seconds**
Minimum	**2.32 seconds**
Maximum	**8.24 seconds**

Question Category	Count	Mean Latency
Answerable	**12**	**4.87 seconds**
Unsupported	**4**	**3.37 seconds**
Follow-up	**4**	**5.32 seconds**

These measurements represent complete chat responses and are distinct from the shorter retrieval/rewrite measurements.

An earlier run returned HTTP 503 for E20 after approximately 72 seconds. E20 succeeded in the selected subsequent run. The failure's cause was not established, so the successful run does not demonstrate that the earlier reliability issue was fixed.

6. Local Memory Evaluation

Method and Scope

A separate Python sampler queried macOS ps for resident set size (RSS), with a 0.5-second wait between readings. RSS values were converted from KiB to MiB.

The sampler monitored:

Process Description	PID
Web server serving requests	65486
Document-processing worker	65480
Django reload supervisor	65479
runlocal launcher	65476

The application total is the simultaneous sum of these four processes. PostgreSQL, the browser, the evaluation runner, the memory sampler, and remote OpenAI infrastructure were excluded.

Workloads

Workload	Observation Window	Samples
Idle	**19.2 seconds**	**37**
Chat evaluation	**115.8 seconds**	**220**
Document processing	**41.7 seconds**	**79**

Chat memory was measured during an additional evaluation run, separately from the selected answer-quality and latency run.

The document-processing workload used background_demo.txt, a 119-byte plain-text file. Finder reported 4 KB allocated on disk; 119 bytes is the file's content size. These windows are observation periods, not measurements of task completion time.

Results (MiB)

Process	Idle (Mean / Peak)	Chat (Mean / Peak)	Processing (Mean / Peak)
Web server	29.50 / 29.73	64.62 / 80.05	59.98 / 79.64
Worker	23.19 / 24.64	23.58 / 24.70	45.83 / 114.91
Reload supervisor	3.64 / 3.64	3.64 / 3.64	3.64 / 3.64
Launcher	4.95 / 4.95	4.94 / 4.95	4.95 / 4.95
Application Total	**61.29 / 62.97**	**96.78 / 113.30**	**114.40 / 203.12**

The worker remained near its idle memory level during chat evaluation and used more memory during document processing. The highest observed application total was 203.12 MiB during processing.

Individual process peaks may occur at different times. Consequently, the peak application total need not equal the sum of individual peaks.

7. Limitations and Interpretation

The selected run met the expected behavior for all 20 cases, with relevant supporting citations for all 16 answerable and follow-up responses. Filtering also rejected matches for the four unrelated questions without losing the expected evidence.

These findings are limited to a small, fixed demonstration dataset and four short documents. They do not establish general chatbot accuracy, production reliability, concurrent-user capacity, or performance on larger document collections. Manual judgments were not independently reviewed by a second evaluator.

Memory tests ran sequentially in the same application session, so later phases may include retained allocations from earlier work. RSS can count shared memory more than once, and sampling may miss brief peaks. The 119-byte processing file does not represent large TXT files, PDFs, scanned documents, or concurrent processing.

The measured results support the demonstrated workflows: evidence-based store answers, appropriate refusal of unrelated questions, follow-up interpretation, and separate background document processing. Broader testing would be needed to assess production workloads.